CASE STUDY 10: FOOD WASTE MONITORING SYSTEM

1. Problem Statement

Food Waste Monitoring System is a simple program used to monitor the amount of food prepared and the amount of food wasted in a canteen.

The program accepts the quantity of food prepared and the quantity of food wasted in kilograms. It calculates the quantity of food consumed and the food waste percentage.

The program also displays a message based on the waste percentage to help identify whether food waste is low or high.

2. Objectives

1. To calculate the quantity of food consumed.
2. To calculate the food waste percentage.
3. To identify food waste levels using conditional statements.
4. To validate the user input.
5. To understand basic Python programming concepts.

3. Assumptions

1. Food quantities are entered in kilograms.
2. The quantity of food prepared must be greater than zero.
3. The quantity of food wasted cannot be negative.
4. Food wasted cannot be greater than food prepared.
5. If the waste percentage is less than or equal to 10%, waste is considered low.
6. If the waste percentage is greater than 10%, waste is considered high.

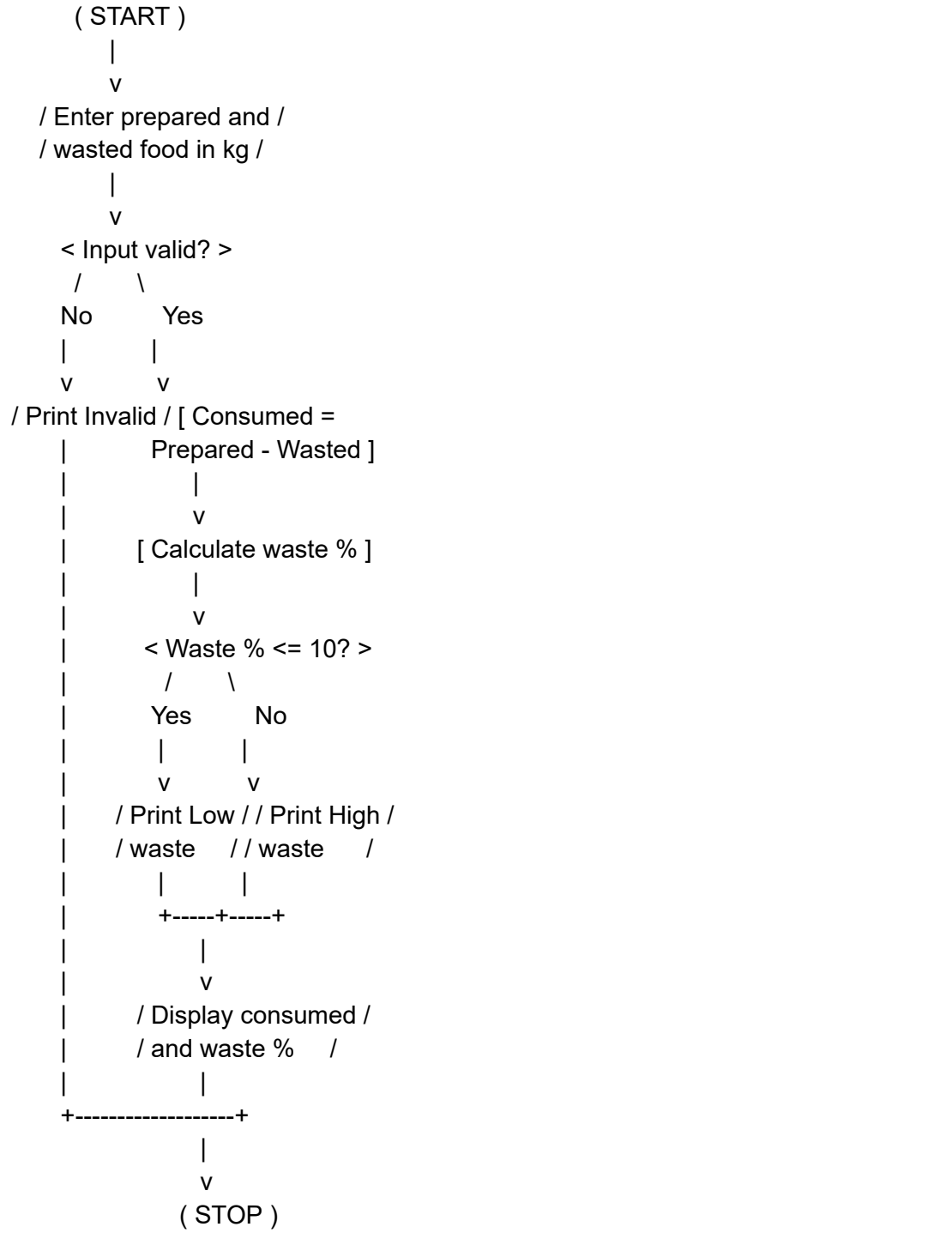
4. Algorithm

1. Start the program.
2. Read the quantity of food prepared.
3. Read the quantity of food wasted.
4. Check whether the input values are valid.
5. If the prepared quantity is less than or equal to zero, display an invalid input message.
6. If the wasted quantity is negative or greater than the prepared quantity, display an invalid input message.
7. Calculate food consumed by subtracting food wasted from food prepared.

8. Calculate the food waste percentage.
9. If the waste percentage is less than or equal to 10%, display “Low food waste”.
10. Otherwise, display “High food waste”.
11. Display the food consumed and waste percentage.
12. Stop the program.

5. Flowchart

Draw the following flowchart using standard symbols in your record.



6. Pseudocode

```
START

INPUT prepared
INPUT wasted

IF prepared <= 0 OR wasted < 0 OR wasted > prepared
   PRINT "Invalid input"
ELSE
   consumed = prepared - wasted
   percentage = (wasted / prepared) * 100

   PRINT consumed
   PRINT percentage

   IF percentage <= 10
      PRINT "Low food waste"
   ELSE
      PRINT "High food waste"
   END IF
END IF

STOP
```

7. Python Source Code

```
print("FOOD WASTE MONITORING SYSTEM")

prepared = float(input("Enter food prepared in kg: "))
wasted = float(input("Enter food wasted in kg: "))

if prepared <= 0 or wasted < 0 or wasted > prepared:
   print("Invalid input")
else:
   consumed = prepared - wasted
   percentage = (wasted / prepared) * 100

   print("Food consumed:", round(consumed, 2), "kg")
   print("Food waste percentage:", round(percentage, 2), "%")

   if percentage <= 10:
      print("Low food waste")
   else:
      print("High food waste")
```

8. Test Cases and Expected Outputs

Test Case 1: Low food waste

Input:

Enter food prepared in kg: 100
Enter food wasted in kg: 5

Expected Output:

FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 100
Enter food wasted in kg: 5
Food consumed: 95.0 kg
Food waste percentage: 5.0 %
Low food waste

Explanation: Out of 100 kg of prepared food, 5 kg is wasted. The waste percentage is 5%, so the program displays “Low food waste”.

Test Case 2: High food waste

Input:

Enter food prepared in kg: 50
Enter food wasted in kg: 15

Expected Output:

FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 50
Enter food wasted in kg: 15
Food consumed: 35.0 kg
Food waste percentage: 30.0 %
High food waste

Explanation: Out of 50 kg of prepared food, 15 kg is wasted. The waste percentage is 30%, so the program displays “High food waste”.

Test Case 3: Invalid food quantity

Input:

Enter food prepared in kg: 20
Enter food wasted in kg: 25

Expected Output:

FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 20
Enter food wasted in kg: 25

Invalid input

Explanation: The wasted food quantity is greater than the prepared food quantity. Therefore, the program displays “Invalid input”.

9. Output Screenshots

```
print("FOOD WASTE MONITORING SYSTEM")

prepared = float(input("Enter food prepared in kg: "))
wasted = float(input("Enter food wasted in kg: "))

if prepared <= 0 or wasted < 0 or wasted > prepared:
    print("Invalid input")
else:
    consumed = prepared - wasted
    percentage = (wasted / prepared) * 100

    print("Food consumed:", round(consumed, 2), "kg")
    print("Food waste percentage:", round(percentage, 2), "%")

    if percentage <= 10:
        print("Low food waste")
    else:
        print("High food waste")
```

```
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case10
.py
FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 100
Enter food wasted in kg: 5
Food consumed: 95.0 kg
Food waste percentage: 5.0 %
Low food waste
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case10
.py
FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 50
Enter food wasted in kg: 15
Food consumed: 35.0 kg
Food waste percentage: 30.0 %
High food waste
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case10
.py
FOOD WASTE MONITORING SYSTEM
Enter food prepared in kg: 20
Enter food wasted in kg: 25
Invalid input
```

10. Conclusion

The Food Waste Monitoring System calculates food consumption and food waste percentage using Python. It validates the input and classifies food waste as low or high based on the calculated percentage. This program demonstrates the use of variables, arithmetic operations, conditional statements, and input validation in solving a real-life problem.